

The Recursive walk() Function

While the match functions check whether a file matches the search criteria, the `walk()` function finds all the files to check. The recursive `walk()` function is passed the name of a base folder to search along with a match function to call for each file in the folder.

The `walk()` function also recursively calls itself for each subfolder in the base folder it's searching. These subfolders become the base folder in the recursive call. Let's ask the three questions about this recursive function:

What is the base case? When the function has finished processing each file and subfolder in its given base folder.

What argument is passed to the recursive function call? The base folder to search and the match function to use for finding matched files. For each subfolder in this folder, a recursive call is made with the subfolder as the new folder argument.

How does this argument become closer to the base case?

Eventually, the function either recursively calls itself on all the subfolders or encounters base folders that don't have any subfolders.

[Figure 10-1](#) shows an example filesystem along with the recursive calls to `walk()`, which it makes with a base folder of `c:\`.



Figure 10-1: An example filesystem and the recursive `walk()` function calls over it

Let's take a look at the `walk()` function's code:

```
def walk(folder, matchFunc):
    """Calls the match function with every file in the folder and its
```